THE DETERMINANT OF A FLAT FINITELY PRESENTED MODULE, DEFINED AS THE ANNIHILATOR INSIDE ITS EXTERIOR ALGEBRA, IS INVERTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The local splitting of a flat module of finite presentation as a direct summand of a finite free module. Invoked at the start of the counted proof (source lines 4769-4771) as a pointer to a lemma of this chapter. Reduces the statement to modules which are direct summands of finite free modules.
- The description of a quasi-coherent module associated to a module over the ring of global sections, and the algebraic description of the determinant of a finite projective module as the annihilator of the module inside its exterior algebra. Both are invoked inside the counted proof (source lines 4778-4787) as pointers to a lemma of this chapter and to results of another chapter. Transfers the algebraic statement to the sheaf level and identifies the two descriptions of the determinant.

Source reference. Original lemma lemma-flat-locally-finite-presentation: sources/stacks-modules/pinned-original-source.tex, source lines 3041–3041. Navigation pointer only: the counted text refers to the Stacks item lemma-flat-locally-finite-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-construct-quasi-coherent-sheaves: sources/stacks-modules/pinned-original-source.tex, source lines 1425–1425. Navigation pointer only: the counted text refers to the Stacks item lemma-construct-quasi-coherent-sheaves, which belongs to another chapter of the project and is not part of this

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Source reference. Original reference more-algebra-section-determinants: sources/stacks-modules/pinned-original-source.tex, source lines 4785–4785. Navigation pointer only: the counted text refers to the Stacks item more-algebra-section-determinants, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-remark-determinant-as-socle: sources/stacks-modules/pinned-original-source.tex, source lines 4786–4786. Navigation pointer only: the counted text refers to the Stacks item more-algebra-remark-determinant-as-socle, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a flat and finitely presented $\mathcal{O}_X$-module. Denote*

$$\det(\mathcal{F}) \subset \wedge^*_{\mathcal{O}_X}(\mathcal{F})$$

the annihilator of $\mathcal{F} \subset \wedge^*_{\mathcal{O}_X}(\mathcal{F})$. Then $\det(\mathcal{F})$ is an invertible $\mathcal{O}_X$-module.

Proof. To prove this we may work locally on X. Hence we may assume $\mathcal{F}$ is a direct summand of a finite free module, see Lemma lemma-flat-locally-finite-presentation. Say $\mathcal{F} \oplus \mathcal{G} = \mathcal{O}_X^{\oplus n}$. Set $R = \mathcal{O}_X(X)$. Then we see $\mathcal{F}(X) \oplus \mathcal{G}(X) = R^{\oplus n}$ and correspondingly $\mathcal{F}(U) \oplus \mathcal{G}(U) = \mathcal{O}_X(U)^{\oplus n}$ for all opens $U \subset X$. We conclude that $\mathcal{F} = \mathcal{F}_M$ as in Lemma lemma-construct-quasi-coherent-sheaves with $M = \mathcal{F}(X)$ a finite projective R-module. In other words, we have $\mathcal{F}(U) = M \otimes_R \mathcal{O}_X(U)$. This implies that $\det(M) \otimes_R \mathcal{O}_X(U) = \det(\mathcal{F}(U))$ for all open $U \subset X$ with det as in More on Algebra, Section more-algebra-section-determinants. By More on Algebra, Remark more-algebra-remark-determinant-as-socle we see that

$$\det(M) \otimes_R \mathcal{O}_X(U) = \det(\mathcal{F}(U)) \subset \wedge^*_{\mathcal{O}_X(U)}(\mathcal{F}(U))$$

is the annihilator of $\mathcal{F}(U)$. We conclude that $\det(\mathcal{F})$ as defined in the statement of the lemma is equal to $\mathcal{F}_{\det(M)}$. Some details omitted; one has to be careful as annihilators cannot be defined as the sheafification of taking annihilators on sections over opens. Thus $\det(\mathcal{F})$ is the pullback of an invertible module and we conclude. □